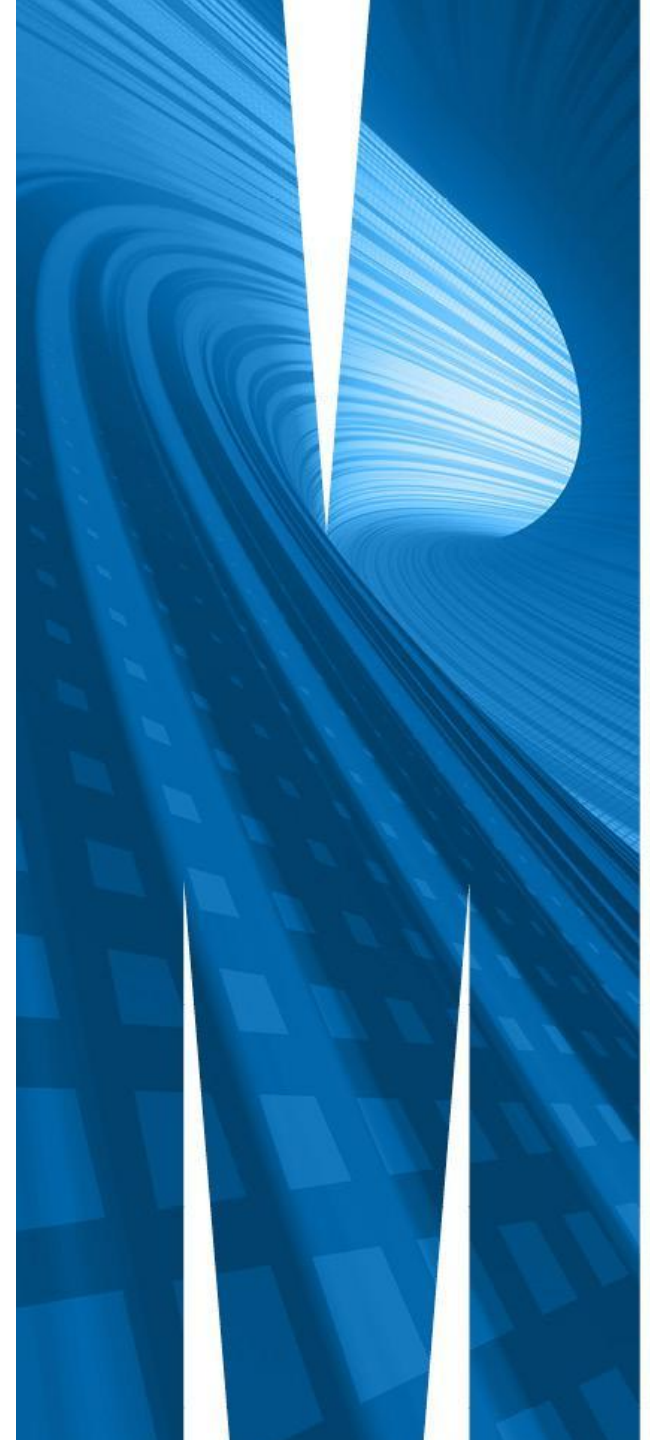


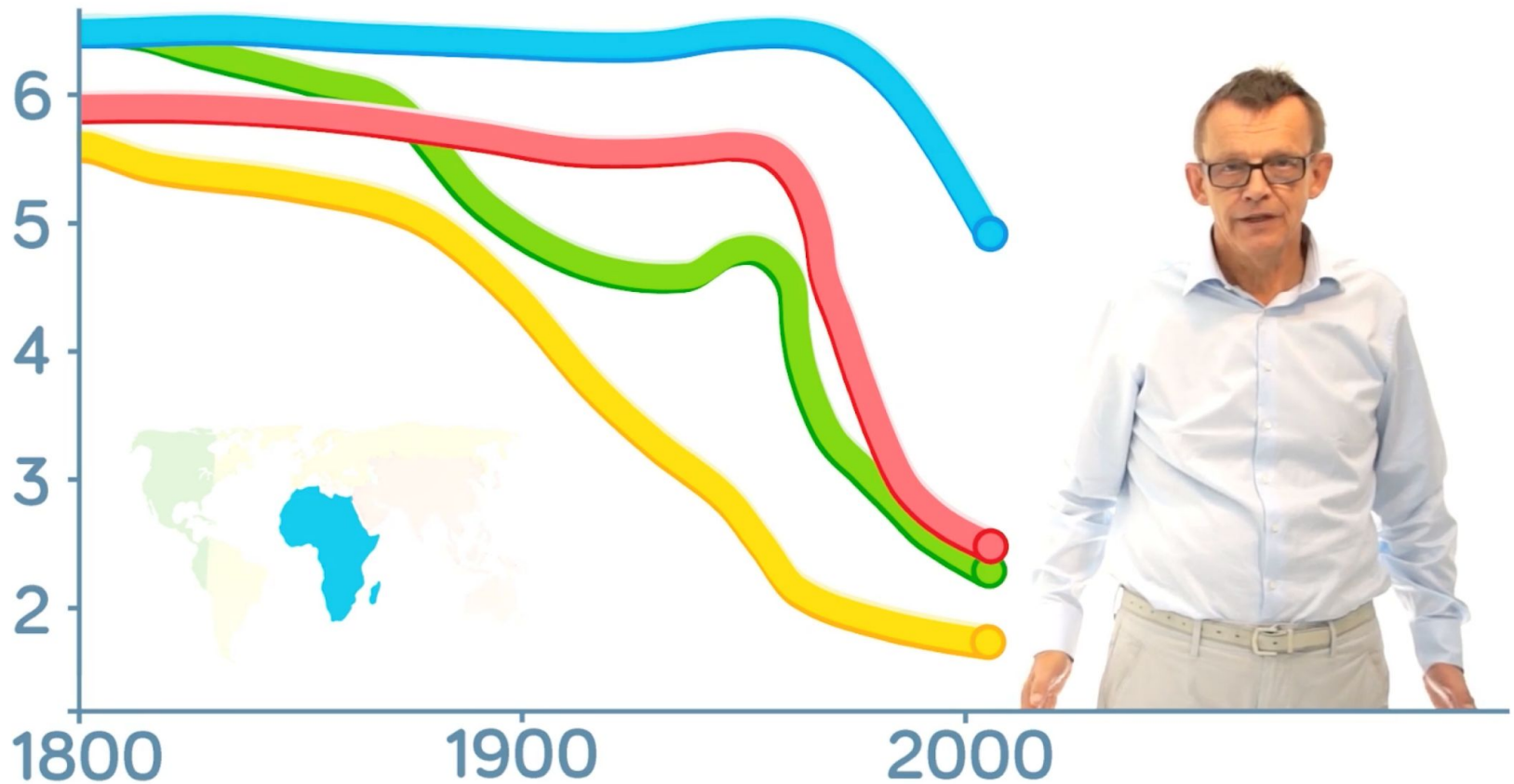
FIT2179 Data Visualisation

Week 10: Animation



- Any questions about the pre-recorded video lectures?
 1. Frames rendered by the browser or encoded in video
 2. Temporal animation: change marks and channels in time
 3. Non-temporal animation: animate between idioms
 4. Cognitive load, short, and long-term memory
 5. Change blindness
 6. Directing user attention with annotation and modifying saliency

Babies per woman



Questions to ask when evaluating animation for data visualisation

What is animated?

- How many items are animated?
- What are the **animated channels and marks**?
- Are **data visualisation principles** applied correctly?
- What **type of change** is the animation showing? For example, movement, processes or algorithms, storytelling, time, zoom for showing relative size.

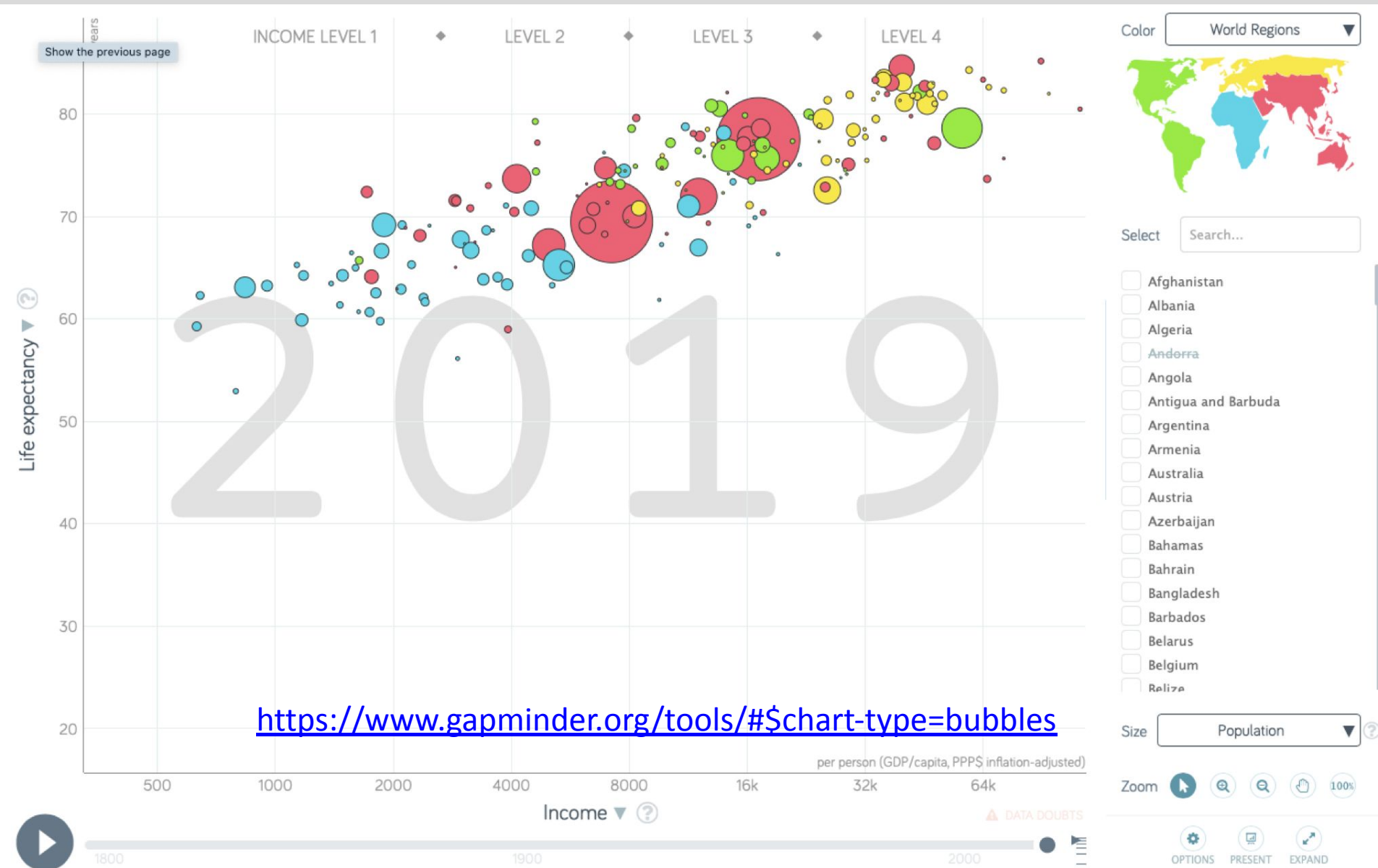
Short-term memory and change blindness

- What techniques are used to **reduce cognitive load**?
- What aspects rely on **short-term working memory**?
- Is **change blindness** an issue?
- Is the visualisation **guiding user attention** and focus?

Speculative assessment

- Do you think...
 - the **main message is made permanent in long term**

Temporal Animation



Questions to ask when evaluating animation for data visualisation

What is animated?

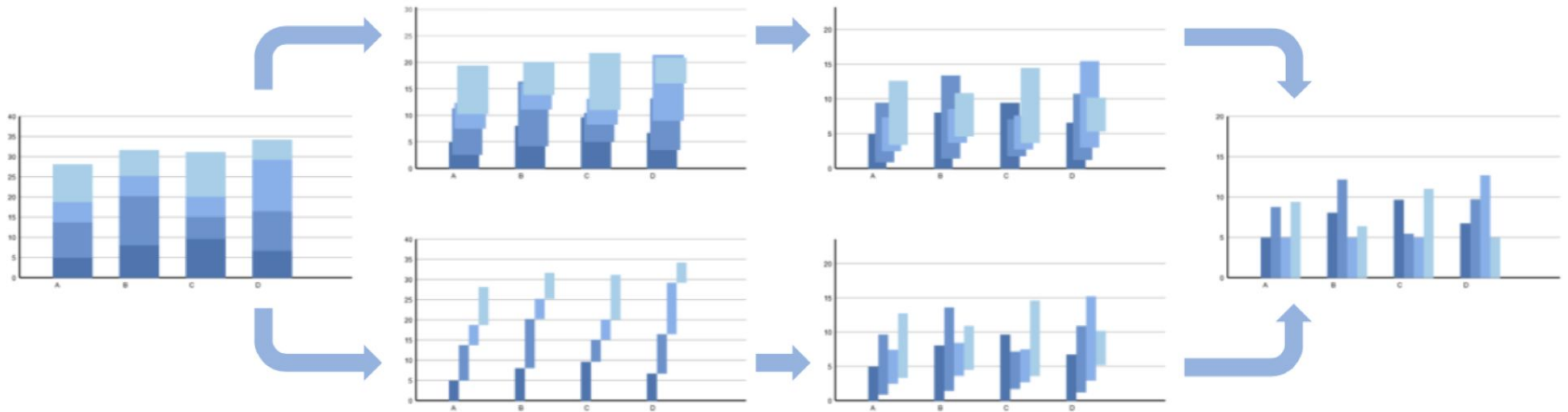
- **How many items** are animated?
- What are the **animated channels and marks**?
- Are **data visualisation principles** applied correctly?
- What **type of change** is the animation showing? For example, movement, processes or algorithms, storytelling, time, zoom for showing relative size.

Short-term memory and change blindness

- What techniques are used to **reduce cognitive load**?
- What aspects rely on **short-term working memory**?
- Is **change blindness** an issue?
- Is the visualisation **guiding user attention** and focus?

Speculative assessment

- Do you think...
 - the **main message is made permanent in long term memory**?
 - the **cognitive load** is too high / high / about right / low?
 - the use of animation is **overall good or bad**?



Heer and Robertson 2007
Animated Transitions in Statistical Data Graphics

<https://www.youtube.com/watch?v=vLk7mlAtEXI>

It would be a poor idea to use animation for complex data analysis or a lookup/locate action.

Animation may be useful for data presentation.

→ Present



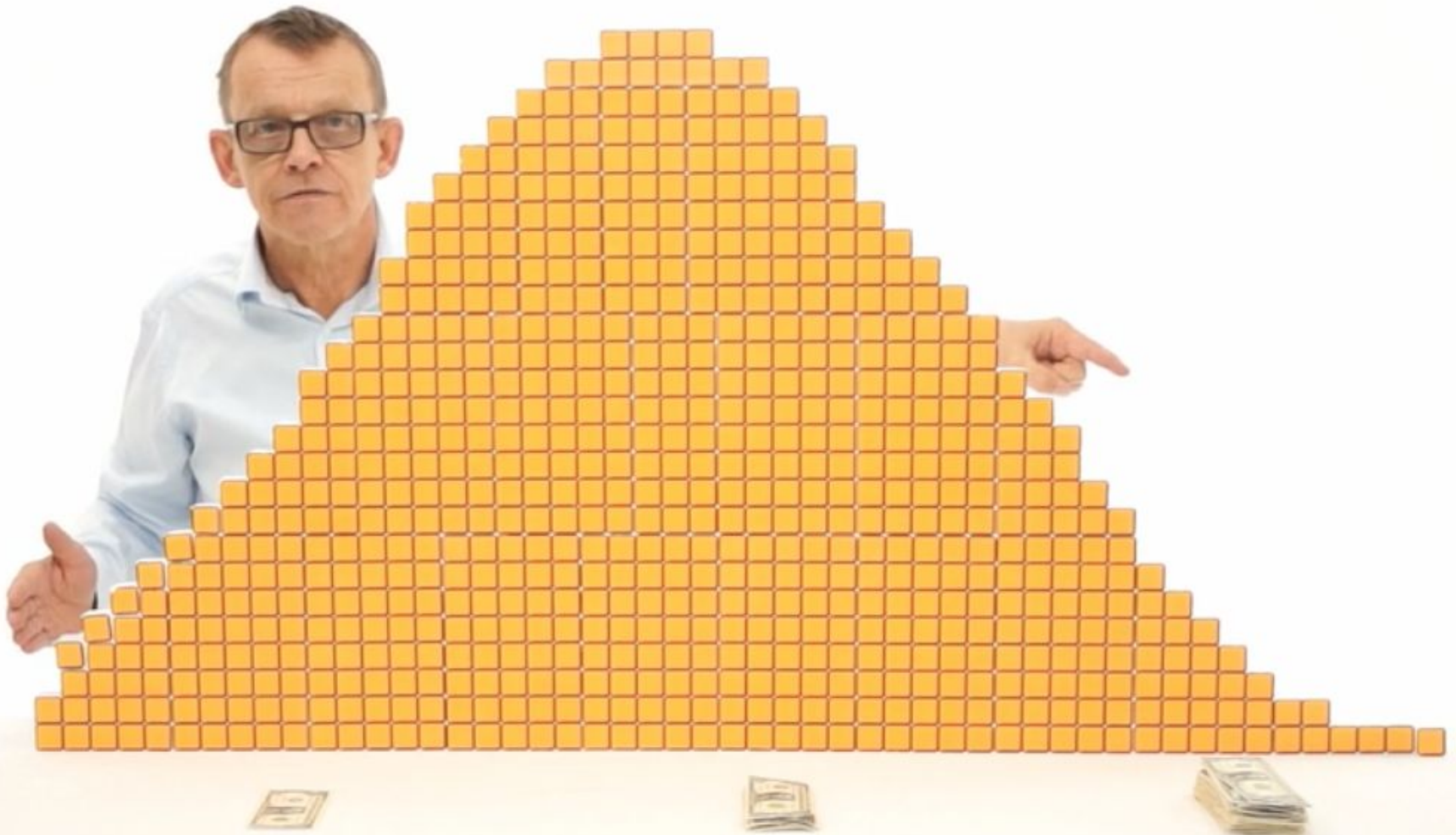
→ Enjoy



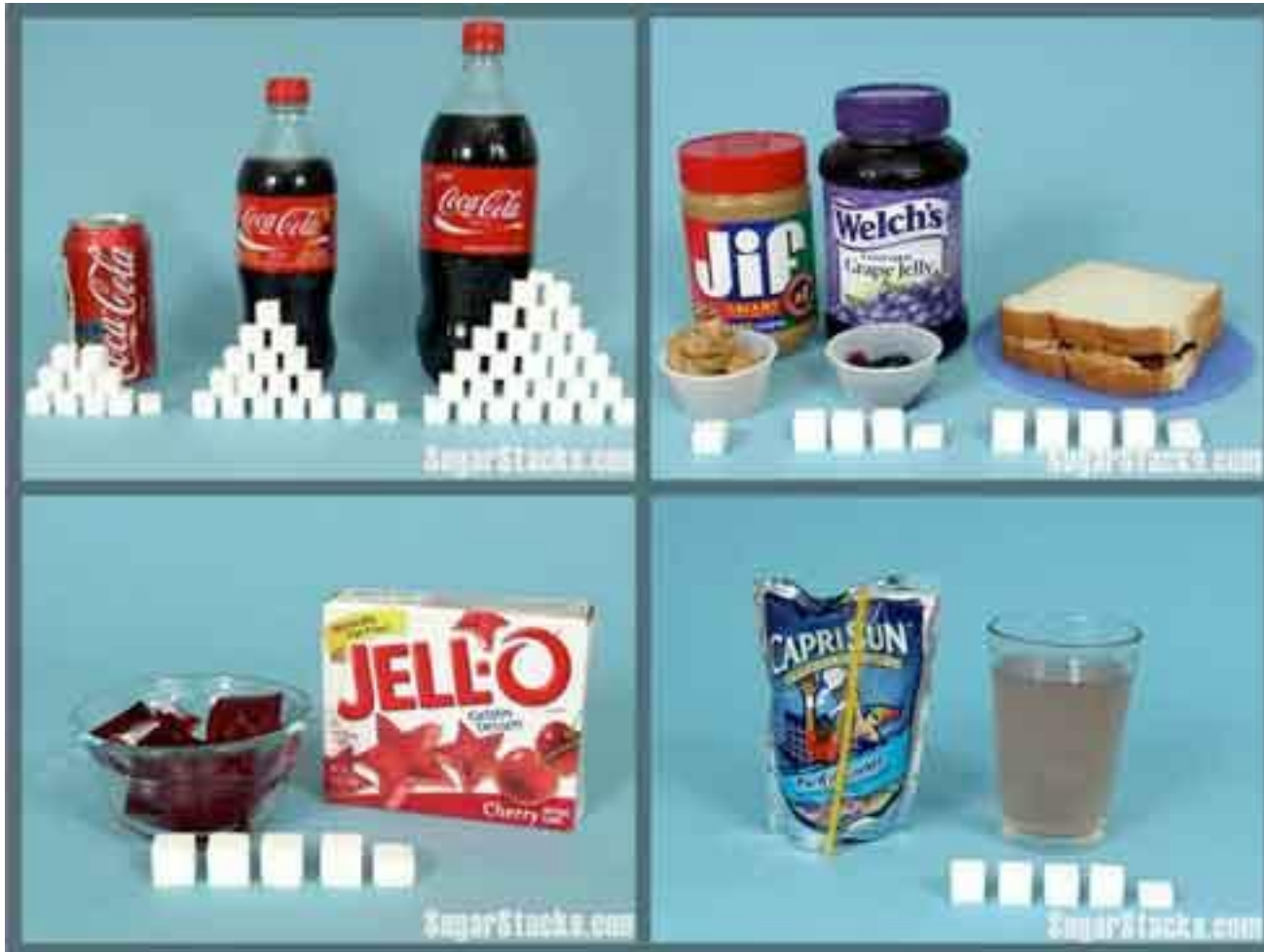
Change Blindness



https://www.youtube.com/watch?v=IGQmdoK_zfY



<https://www.gapminder.org/answers/how-many-are-rich-and-how-many-are-poor/>



This is a chronological list of physical visualizations and related artifacts, maintained by [Pierre Dragicevic](#) and [Yvonne Jansen](#). Thanks to [our contributors](#). Feel free to post a [general comment](#) or if you know of another interesting physical visualization, please [submit it!](#)

This list currently has 351 entries. [See recent additions](#). You can also get notified of new entries through [Twitter](#).



≡ List view

Passive physical visualizations (204)

Active physical visualizations (39)

Physical models (34)

Measuring instruments (11)

Interactive installations (7)

Enabling technologies (26)

Other (7)

Uncertain (23)

dataphys.org/list/gallery/



5500 BC - Mesopotamian Clay Tokens



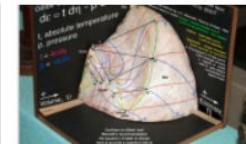
2600 BC - Inca Quipus



500 BC - Visualizing Opinions with Pebbles



1862 - Marshall Islands Stick Charts



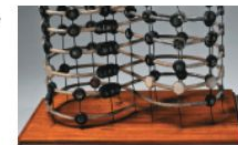
1871 - Thermodynamic Surfaces



1890 - Polynesian Genealogical Instrument



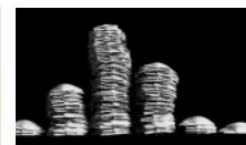
1898 - Tallies Used as Social Displays on Pacific...



1898 - Crookes' Vis Generatrix



1900 - Pearson and Lee's Height Correlation...



1901 - Davenport's Physical Distributions



1907 - Pin Maps



1913 - Frankfurt Streetcar Load



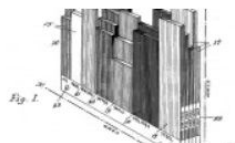
1914 - Solid 3D Curves for Engineering



1915 - Wire Models of Factory Worker Movements



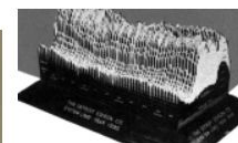
1920 - Yakama Time Ball



1926 - Karsten's Tridimensional Chart



1933 - IBM's Cosmograph



1935 - 3D Visualizations of Power Consumption



1939 - Map of Great Britain's Marine Trade



1940s - Stedman's 3D Periodic Table